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RESEARCH INTERESTS

Low-dimensional topology, in particular braids, knots and links, cobordisms, knot concordance, knotted surfaces, Khovanov and Heegaard Floer homology, 3- and 4-manifolds.

EMPLOYMENT

12/2025 – 11/2027	Postdoctoral fellow , University of Glasgow , Glasgow, UK. Supported by SNSF Postdoc.Mobility fellowship 230329.
10/2023 – 11/2025	Postdoctoral fellow , Max Planck Institute for Mathematics , Bonn, Germany. Mentor: Arunima Ray.
07/2019 – 07/2023	Scientific assistant , ETH Zurich , Zurich, Switzerland.
10/2018 – 04/2019	Teaching assistant , Ulm University , Ulm, Germany.
08/2017 – 05/2018	Teaching assistant , Syracuse University , Syracuse, NY, USA.
10/2013 – 07/2017	Teaching assistant , Ulm University , Ulm, Germany.

EDUCATION

07/2019 – 06/2023	Doctor of Science in Mathematics, ETH Zurich , Zurich, Switzerland. Thesis: <i>On notions of braid positivity and knot concordance</i> . Advisor: Peter Feller.
08/2016 – 02/2019	Master of Science in Mathematics (1.0, with distinction), Ulm University , Ulm, Germany. Minor: Computer Science. Thesis: <i>Massey products and linking numbers</i> . Supervisor: Stefan Wewers.
08/2017 – 05/2018	Master of Science in Mathematics (GPA 4.0), Syracuse University , Syracuse, NY, USA.
10/2012 – 08/2016	Bachelor of Science in Mathematics (1.5), Ulm University , Ulm, Germany. Minor: Physics. Thesis: <i>Das Existenzproblem von Hurwitz</i> . Supervisor: Stefan Wewers.
09/2014 – 02/2015	Visiting semester , ETH Zurich , Zurich, Switzerland.

SELECTED AWARDS

12/2025 – 11/2027	Swiss National Science Foundation Postdoc.Mobility fellowship . <i>Project 230329: Concordance of braid positive knots</i> .
2017	Fulbright Travel Grant .
2014 – 2017	Talanx Scholarship .
2013 – 2014	Deutschlandstipendium Scholarship (Ulm University, EY Foundation).
2012 – 2013	Deutschlandstipendium Scholarship (Ulm University, Rotary Club Ulm).

RESEARCH SEMESTER

Participant in the **AIM Online Research Community *4-Dimensional Topology*** organized by Miriam Kuzbary, Maggie Miller, Juanita Pinzón-Caicedo and Hannah Schwartz in Spring 2021.

PREPRINTS

1. **On the detection of knotted spheres by their traces in high dimensions**, with V. Bais, A. Di Prisa, D. Hartman, C.-S. Hsueh, M. Kegel, A. Merz, M. Pencovitch, A. Ray, D. Santoro and L. Wakelin. [ArXiv:2511.07251](#).

PUBLISHED AND ACCEPTED PAPERS (IN REVERSE ORDER OF WRITING)

8. **Braid positive surgery diagrams**, with M. Kegel.
Glasg. Math. J. (2026) pp. 1-6. DOI: 10.1017/S0017089526100913. [ArXiv:2512.14589](#).
7. **Non-complex cobordisms between quasipositive knots**, with M. Borodzik.
J. Math. Pures Appl. (9) 207:11 (2026). Id/No 103842. [ArXiv:2504.04894](#).
6. **Algorithms in 4-manifold topology**, with S. Bastl, R. Burke, R. Chatterjee, S. Dey, A. Durst, S. Friedl, D. Galvin, A. García Rivas, T. Hirsch, C. Hobohm, C.-S. Hsueh, M. Kegel, F. Kern, S. M. S. Lee, C. Löh, N. Manikandan, L. Mousseau, L. Munser, M. Pencovitch, P. Perras, M. Powell, J. P. Quintanilha, L. Schambeck, D. Suchodoll, M. Tancer, A. Thiele, M. Uschold, S. Veselá, M. Weiß and M. von Wunsch-Rolshoven.
Accepted for publication in *Algebr. Geom. Topol.*. [ArXiv:2411.08775](#).
5. **Minimal cobordisms between thin and thick torus knots**, with S. Baader, L. Lewark and F. Misev. *Proc. Edinb. Math. Soc., II. Ser.* 69, no. 1 (2026), pp. 177–183. [ArXiv:2405.13719](#).
4. **3-braid knots with maximal 4-genus**, with S. Baader, L. Lewark and F. Misev.
Trans. Amer. Math. Soc. Ser. B 11 (2024), pp. 600–621. [ArXiv:2303.11918](#).
3. **Strongly quasipositive links are concordant to infinitely many strongly quasipositive links**. Accepted for publication in *Proc. Amer. Math. Soc.* [ArXiv:2210.06612](#).
2. **On the nonorientable four-ball genus of torus knots**, with F. Binns, S. Kang and J. Simone.
Algebr. Geom. Topol. 25-4 (2025), pp. 2209–2251. [ArXiv:2109.09187](#).
1. **The upsilon invariant at 1 of 3-braid knots**.
Algebr. Geom. Topol. 23-8 (2023), pp. 3763–3804. [ArXiv:2109.09187](#).

CONFERENCE PROCEEDINGS (NOT PEER-REVIEWED)

5. **MATRIX-MFO Tandem Workshop: Invariants in Low-Dimensional Toplogy. Combinatorics, Geometry, and Computation**, with S. Friedl, J. Licata, S. Tillmann. *Oberwolfach Rep.* 21 (2024), no. 3, pp. 2511–2550.
4. **Obstructing ribbon clasp number one**, with D. Celoria, A. Di Prisa, J. Howie, J. Licata, F. Misev, L. Schambeck, L. Wakelin. *Oberwolfach Rep.* 21 (2024), no. 3, pp. 2511–2550.
3. **Notions of braid positivity and knot concordance**. *Oberwolfach Rep.* 21 (2024), no. 3, pp. 2511–2550.
2. **Strongly quasipositive knots are concordant to infinitely many strongly quasipositive knots**. *Oberwolfach Rep.* 20 (2023), no. 1, pp. 172–174.
1. **Concordance of positive braid knots**. *Oberwolfach Rep.* 18 (2021), no. 3, pp. 2360–2362.

INVITED TALKS AT CONFERENCES AND WORKSHOPS

07/2026*	Young Women in Geometric Topology , Bonn.
06/2026*	Berlin-Brandenburg Workshop VII: Knot theory and its applications, Berlin.
10/2025	Trisections and related topics , Marseille.
11/2024	IberoSing International Workshop 2024: Low-dimensional Topology & Singularity Theory , Madrid.
07/2024	Mini-symposium <i>Knot theory and low dimensional manifolds</i> , European Congress of Mathematics , Seville.
04/2024	NRW Topology Meeting , Wuppertal.
04/2024	Manifolds and groups in Bologna II , Bologna.
09/2023	Leaning into Topology Workshop , Pisa.
09/2023	Swiss Knots , Regensburg.
06/2023	Mini Workshop, Bern.
03/2023	The low dimensional workshop , Budapest,  .
11/2022	Paroles aux jeunes chercheuses et chercheurs en géométrie et dynamique , Orsay.
09/2022	Mini-symposium <i>Algebra and Low-Dimensional Topology</i> , DMV Annual Meeting 2022 , Berlin.
06/2022	Surfaces in 4-manifolds , Le Croisic.

INVITED TALKS AT SEMINARS

02/2026	Geometry and Topology Seminar , Glasgow.
01/2026	Geometry Seminar , MPI MiS Leipzig.
10/2025	Algebra and Geometry Seminar , Neuchâtel.
07/2025	Oberseminar Global Analysis and Geometry , Regensburg.
06/2025	Séminaire Géométrie des Espaces Singuliers , Lille.
05/2025	Geometry seminar , ETH Zurich.
12/2024	Oberseminar Topologie , University of Bonn.
09/2024	Séminaire de Géométrie, Algèbre, Dynamique et Topologie , Dijon.
05/2024	Séminaire de Géométrie et Topologie de Marseille , Marseille.
05/2024	Oberseminar Topologie , Münster.
04/2024	Geometry & Topology Seminar , Glasgow.
03/2024	MPIM Oberseminar , MPIM Bonn.
11/2023	MPIM Topology Seminar , MPIM Bonn, short talk.
05/2023	Low-dimensional topology seminar , MPIM Bonn.
12/2022	Geometry and Topology Seminar , CIRGET, Montréal, online,  .
06/2022	LKS-Seminar , Regensburg.
05/2022	Geometry & Topology Seminar , Caltech, online.
05/2022	Oberseminar Geometrie , Fribourg.
02/2022	Knot Online Seminar (K-OS) , online,  .
04/2021	AIM Online Research Community 4-Dimensional Topology , lightning talk.

INVITED TALKS AT GRADUATE COLLOQUIA

12/2023	Pitt AWM Student Seminar , Pittsburgh, online.
10/2023	Geometry Graduate Colloquium , ETH Zurich.
10/2022	Bern–Fribourg Graduate Seminar , Bern.
05/2022	Baby Geometri Seminar , Pisa.
04/2022	Bernoullis Tafelrunde , Basel.
11/2021	Zurich Graduate Colloquium , Zurich.
04/2021	Geometric Topology Grad and Postdoc Seminar (GT GAPS) , online.

CONTRIBUTED TALKS AT CONFERENCES AND WORKSHOPS

09/2024	MATRIX-MFO Tandem Workshop: Invariants in Low-Dimensional Topology. Combinatorics, Geometry, and Computation , MFO Oberwolfach, short talk.
02/2024	BTW4: Knot Floer Homology , Besse.
02/2023	Winter Braids XII , Tours.
01/2023	Low dimensional topology , MFO Oberwolfach.
05/2022	Swiss-Mathematical-Society Doctoral Day 2022 , Fribourg.
04/2022	BTW2: Sliceness, exotic pairs, and quantum invariants , Aussois.
09/2021	Invariants and Structures in Low-Dimensional Topology , MFO Oberwolfach, short talk.

OTHER CONFERENCES AND WORKSHOPS ATTENDED

07/2026*	Low-dimensional Topology , MFO Oberwolfach.
06/2026*	Geometry and Topology in Low Dimensions: Lasting Trends and Emerging Directions , Budapest.
12/2025	Workshop on surfaces in 4-manifolds , Regensburg.
10/2025	36. NRW Topology Meeting , Bonn.
07/2025	Categorification in Low Dimensional Topology , Bochum.
06/2025	Swiss Knots , Geneva.
09/2024	Workshop on 4-manifolds and algorithms , Regensburg.
02/2024	Winter Braids XIII , Montpellier.
09/2023	GRAZP: Groups and Rigidity Around the Zimmer Program , Ventotene.
07/2023	Gauge Theory and Topology: in Celebration of Peter Kronheimer's 60th Birthday , Oxford.
01/2023	Winter school on singularities and low-dimensional topology , Budapest.
08/2022	Frontiers in Geometry and Topology Summer School , Trieste.
04/2022	Braids in Low-Dimensional Topology , ICERM, Providence.
08/2021	Perspectives on quantum link homology theories , Universität Regensburg.
07/2021	Tech Topology Summer School , Georgia Institute of Technology, online.
07/2021	Swiss Knots , Fribourg.
06/2021	NCNGT: Nearly Carbon Neutral Geometric Topology Conference , online.
06/2020	NCNGT: Nearly Carbon Neutral Geometric Topology Conference , online.
02/2020	Winter Braids X , Pisa.
10/2019	Regional workshop on foliations and 3-manifolds , Universität Regensburg.
10/2019	Low-dimensional topology workshop , Universität Regensburg.
09/2019	Workshop on 4-manifolds , Max Planck Institute for Mathematics Bonn.
09/2019	Of coarse! Quasi-isometries and groups: rigidity and classification , Ventotene.
07/2019	Swiss Knots , ETH Zurich.
10/2017	27th annual Route 81 conference on commutative algebra and algebraic geometry , Cornell University, Ithaca, NY, USA.

* means planned.

SUPERVISION

- 02/2025 – 10/2025 **Master thesis** by Chiara Donatone at Università degli Studi di Milano.
Thesis title: *On RBG links and knot traces*.
Co-supervision with Diego Matessi.
Will continue as PhD student at the University of Glasgow.
- 06/2024 – 06/2025 **Master thesis** by Frieda Kern at the University of Bonn.
Thesis title: *Linking number obstructions to satellite homomorphisms*.
Co-supervision with Peter Teichner.
Continues as [PhD student](#) at the University of Glasgow.

TEACHING

- Winter 2024/25 **University of Bonn/MPIM Bonn**
Lecturer for [Topics in 3-manifold topology](#) (Master in Mathematics).
- Spring 2022 **ETH Zurich**
Course organizer for **Analysis II** (Bachelor in Civil, Geospatial, or Environmental Engineering).
- Fall 2021 Course organizer for **Analysis I** (Bachelor in Civil, Geospatial, or Environmental Engineering).
- Spring 2021 Course organizer for [Topologie](#) (Bachelor in Mathematics).
- Fall 2020 Course organizer for [Lineare Algebra I](#) (Bachelor in Mathematics, Physics, or Interdisciplinary Sciences).
- Spring 2020 Course organizer for [Algebraic Topology II](#) (Bachelor or Master in Mathematics).
- Fall 2019 Teaching assistant for [Complex Analysis](#) (Bachelor in Mathematics, Physics, or Interdisciplinary Sciences).
- Spring 2018 **Syracuse University**
Teaching assistant for **Calculus III**.
- Fall 2017 Teaching assistant for **Calculus III**.
- Summer 2019 **Ulm University**
Instructor for **Linear Algebra Review Course** (Bachelor in Computer Science and Engineering).
- Winter 2013/14 – Summer 2019 Student teaching assistant for **Analysis I-III** and **Linear Algebra I-II**.

TEACHER TRAINING

- Winter 2024/25 Participation in the [Teaching Mentoring Program](#), Hausdorff School for Mathematics in Bonn.
Mentors: Viktoriya Ozornova and Arunima Ray (both MPIM Bonn).
- 09/2021 Certificate **Learning to Teach**.
Program for doctoral teaching assistants, ETH Zurich.
Teaching project: *How to teach writing proofs—classroom assessment techniques in mathematics classes*.

OUTREACH ACTIVITIES

03/2026	Co-organizer of GROW@Dresden 2026 (Graduate Research Opportunities for Women at Dresden 2026), March 26-27. Joint with Anita Behme, Türkü Özlüm Çelik, Heather Harrington, Oana Padurariu, Andreas Thom.
03/2025	Invited panelist for panel discussion “ <i>What is mathematics research like?</i> ” at GROW@Frankfurt 2025 .
12/2024	Invited speaker at Tea Time with Women in Mathematics , Bonn.
04/2024	Volunteer at the mentoring lunch at GROW@Bonn 2024 .
03/2023	<i>Work as a PhD Student</i> , invited short talk, MindPhair 2023 , ETH Zurich (job-fair for mathematicians, physicists and computational scientists).
03/2022	<i>Work as a PhD Student</i> , invited short talk, MindPhair 2022 , ETH Zurich.

CONFERENCE AND SEMINAR ORGANIZATION

02/2026 –	ArXiv seminar , University of Glasgow.
10/2023 – 11/2025	Low-dimensional topology seminar , MPIM Bonn.
09/2024	MATRIX-MFO Tandem Workshop: Invariants in Low-Dimensional Topology. Combinatorics, Geometry, and Computation , Mathematisches Forschungsinstitut Oberwolfach, Sept 22–27. Joint with Stefan Friedl, Joan Licata and Stephan Tillmann.
09/2020 – 07/2023	Geometry Graduate Colloquium , ETH Zurich, funded by the Zurich Graduate School in Mathematics.

OTHER ACADEMIC SERVICE AND ACTIVITIES

	MPIM Bonn
09/2024	Member of the Postdoc Representative Selection Committee.
	ETH Zurich
09/2021 – 07/2023	Co-organizer of get-togethers for women in math at ETH Zurich and the University of Zurich, funded by the Zurich Graduate School in Mathematics.
08/2021 – 07/2023	Elected member of the Nachdiplom Lectures Committee at the Department of Mathematics.
05/2020 – 05/2023	Member of the Board of VMM (Association of the Mid-level Academic Faculty at the Department of Mathematics).
10/2022	Co-organizer of the first D-MATH (Post)Doctoral Welcome Day.
05/2021 – 05/2022	Mid-level faculty representative in the Department Conference.
	Ulm University
10/2016 – 07/2017	Member of the Mathematics Student Council (Fachschaft Mathematik) at the Department of Mathematics and Economics.

REFEREING AND REVIEWING

2024 – present	Referee and quick opinions for <i>Algebr. Geom. Topol.</i> , <i>Bull. Lond. Math. Soc.</i> and <i>Pac. J. Math.</i>
2024 – present	Reviewer for <i>zbMATH Open</i> .

REFERENCES

[Peter Feller](#), University of Neuchâtel,
[Matthew Hedden](#), Michigan State University,
[Lukas Lewark](#), ETH Zurich,
[Viktoriya Ozornova](#), MPIM Bonn (Teaching),
[Arunima Ray](#), University of Melbourne,

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